# How Does the Anthropomorphism of Automated Vehicles Shape Human Trust? A Systematic Review and Meta-Analysis

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## Abstract

Trust is a critical factor in the acceptance of automated vehicles (AVs), but market data indicate that public trust in AVs remains suboptimal. Of all tested strategies, anthropomorphizing AVs in different modalities, such as adding human-like visual features or human voices, has gained attention. This meta-analysis addresses the role of anthropomorphism in AV trust as a timely response to the ongoing debate. Following PRISMA guidelines, a total of 66 articles and 70 effect sizes, from 5903 participants across seven databases, were included.

The results revealed that anthropomorphic interfaces, compared to non-anthropomorphic interfaces, do not significantly improve trust in AVs (Standardized Mean Difference (SMD) = 0.04, not significant). However, in general, adding auditory anthropomorphic features is recommended for increasing trust compared to visual anthropomorphic features (SMD: 0.30 vs. -0.48). Moreover, different levels of anthropomorphism (superficial vs. deep) did not have a significant impact on trust in AVs. The overall trend indicated that with increasing age of participants, the reliance on anthropomorphic interfaces for increasing trust became more pronounced.

This meta-analysis shows that adding human-like features is not a dependable way to build trust in AVs. Future works are needed on how it can be compared to other approaches like explainability. Generally, as a suggestion for stakeholders, it is important to recognize that the impact of anthropomorphic interfaces on trust in AVs is not the same as assumptions.

**Keywords:** Human-like; Human-AV interaction; Interface design; User Experience, Social Cars; Automated driving; Autonomous vehicles

## 1. Introduction

A review of previously published articles and broadcasted news (Penmetsa et al., 2023), particularly in recent years, predicted that the number of consumers/users of automated vehicles (AVs) would increase annually (Lohaus et al., 2024; O'Hern & St. Louis, 2023; Shin et al., 2022). These predictions were consistent with the increase in the number of AVs on the streets (Agrawal et al., 2023; Norman-López et al., 2026). According to Statista, the global stock of vehicles equipped with at least some level of automation amounted to approximately 31 million in 2019 and is exceeded 54 million by 2024, while the AV market is expected to reach 41.1 billion US Dollars, with an estimated 58 million AVs sold worldwide by 2030 (Carlier, Apr 2, 2025). Although this increment in the statistics cannot translate to trust to AVs and it is still a major concern. According to the American Automobile Association (AAA), in 2021, 14% of American drivers reported trusting AVs, while 54% expressed fear and distrust toward these vehicles. By March 2024, trust levels had declined to 9%, with 66% of drivers reporting fear of AVs. Although there was a slight improvement in February 2025 with 13% expressing trust and 60% still expressing fear, the overall picture remains largely unchanged. Despite some fluctuations over the years, distrust toward self-driving cars continues to significantly outweigh trust, indicating persistent public apprehension (Moye, 2025). Similarly, the J.D. Power Mobility Confidence Index for 2023 reflected a trust score of just 37 out of 100, marking a decrease from 39 in 2022 (J.D.POWER, 2023).

Various conceptual models have introduced trust in automated system such as Hoff and Bashir's three-layer model (Hoff & Bashir, 2015), Chuan Hu et al. framework of the dynamic evolution of trust (C. Hu et al., 2024), and Zhou trust model of pedestrians to AVs (Zhou et al., 2023). Although trust in AVs is characterized as a multidimensional and dynamic concept (Dirsehan & Can, 2020), it encompasses two main aspects, initial trust and dynamic trust (Kraus et al., 2021; Zhou et al., 2023). The first pertains to pre-purchase and pre-usage, indicating individuals' perception and belief in the success of this technology in achieving its objectives, named “Initial Trust” and the second aspect relates to trust during using and/or post usage of an AV, requiring a certain level of trust in its performance, named “Dynamic Trust”. For initial trust, most work sought individuals' subjective perceptions without having an interaction (e.g., (Wu et al., 2023b)). In the dynamic trust, individuals have experienced an interaction with an AV or a simulation of it (e.g., (Lohaus et al., 2024)). Although the role of initial trust in development dynamic trust cannot be ignored (Manchon et al., 2023), numerous other parameters also influence dynamic trust. Most interventions for improving trust have

focused primarily on improving vehicle technical functions, such as sensor accuracy, cameras, and so forth (Cunha et al., 2025). Such technical improvements align with trust improvement models, which indicate that enhancing technical aspects is an effective strategy (Hancock et al., 2023; Siau & Wang, 2018).

Beyond technical improvements, trust can also be improved through Human Factors approaches. A recently introduced point of view, especially following the initial success of anthropomorphism in human-robot interaction, highlights the increasing use of anthropomorphic interfaces in AVs (Saeedeh Mosaferchi, 2025; Sheridan, 2020). The term "anthropomorphism" originates from two distinct Greek roots: "Anthropos" meaning "human" and "Morphe" meaning "shape" or "form" (Ding et al., 2022; Zhang & Sheng, 2026).
Anthropomorphism is attribution of human traits or emotions to non-human entities, such as animals or objects including technologies such as social robots or AVs (Song et al., 2026; Wu et al., 2023a; Wu et al., 2024). Therefore, anthropomorphism can involve designing AVs to mimic human-like behaviors or appearances. With regards to dynamic trust, anthropomorphism assumes to make an AV more familiar to humans (Lawson-Guidigbe, Amokrane-Ferka, et al., 2023a). Additionally, anthropomorphism can evoke emotional responses in users, fostering a sense of connection and empathy towards the AVs (Park et al., 2023). Anthropomorphism's role in increasing the trust of drivers/passengers/pedestrians in an AV (dynamic trust) has been addressed in several studies (Park et al., 2024b; Wu et al., 2023b). Participants were more likely to trust the AV's competence as it incorporated anthropomorphic features (Waytz et al., 2014). A combination of symbolic (normal arrows without any anthropomorphic characteristic) and anthropomorphic information increased trust in AVs compared to symbolic information alone. The authors implied that anthropomorphizing information may foster the perception of AVs as social agents and enhance trust in those AVs (Niu et al., 2018).
Supported by the triadic theory of anthropomorphism (Epley et al., 2007), anthropomorphizing technology can increase trust. This theory suggests that attributing human-like qualities such as mind, intent, or emotions to non-human agents like technology can alter our perception and interaction with them (Troshani et al., 2021). Despite positive results, these have not always been consistent, and in some cases, anthropomorphic interfaces even led to a reduction in trust towards AVs (Messer et al., 2024; P. Wang et al., 2021). When examining trust in AVs, studies have not always contrasted anthropomorphic with non-anthropomorphic designs; rather, they have sometimes compared different levels of anthropomorphism, - a low level and a high level- also referred to as superficial and deep anthropomorphism (Wu et al., 2023b). This gradation

of anthropomorphism has been implemented through various approaches. For instance, some studies have shown that a female voice is perceived as more anthropomorphic than a male voice, attributing this to the perceived warmth of the voice (Deng et al., 2024; Dong et al., 2020; Haziraj, 2025). Similarly, polite vocal tone is sometimes judged to be more human-like than a harsh one (Lee & Lee, 2022; Ribino, 2023), and the addition of a human voice to a visual anthropomorphic feature (Thimm et al., 2024) or a human voice to a human-like text script (Schroeder & Epley, 2016) may increase the perceived depth of anthropomorphism. In this regard, in a recently published article, the differences between superficial and deep anthropomorphizing to AVs is discussed (Wu et al., 2023b). The authors pointed out that a superficial anthropomorphism can significantly boost human trust via mediating role of interaction quality and facilitating condition. Also, they declared that a deep level of anthropomorphism would decrease human trust. Decreasing trust due to anthropomorphizing a car is not concluded to this article and there are other numerous articles in the literature devoted to this adverse effect. Recent research highlights that while anthropomorphic design, such as giving AVs baby-faced features, can increase trust in low automation levels, it may backfire in higher automation level (L5). In high-autonomy contexts, such designs create a mismatch between the AV's expected competence and its naive appearance, leading to higher responsibility attribution in case of accidents. This "cuteness paradox" suggests that human-like designs can unintentionally undermine trust and amplify blame in AV failures (Messer et al., 2024).

Trust in automated vehicles is therefore not only a technical challenge but also broader human factors issue that shapes public acceptance and safe deployment. Some studies report that anthropomorphic features enhance trust, while others suggest that they may decrease trust or even create unintended effects such as the "cuteness paradox" or the uncanny valley. This lack of clarity underscores the need for a systematic synthesis of the literature to evaluate the role of anthropomorphic design in shaping trust toward AVs. Accordingly, the present study aims to conduct a systematic review and meta-analysis to resolve these inconsistencies, clarify the mechanisms linking anthropomorphism and trust, and provide evidence-based insights for the design of safer and more socially acceptable automated vehicles.

In the next section, a summary of important concepts and theories related to the topic are presented, followed by the rationale of the current study. The methodology is presented in the third section, and in the final sections, the results, discussion and practical conclusions are provided.

## 2. Theories and related concepts

### *2.1. Anthropomorphism in human-robot interactions*

In a robot, anthropomorphism means that humans attribute a feature that is generally related to humans to a social robot (e.g., smiling face) (Ruiz-Equihua et al., 2023; Tan et al., 2025). The reason for choosing the robot among other technologies here is the agreement of previous studies on the greater similarity of the concept of anthropomorphism in the AVs to the social robots compared to other technologies (Reuter et al., 2025). That is, researchers believe that attributing a human characteristic to a car is derived from human interaction with robots (AV can be considered a social robot but in motion) (Pelikan & Jung, 2023; Rosa et al., 2020).

Researchers in human-robot interaction have pointed to various human characteristics that have been attributed to robots (appearance vs. thoughts vs. emotions), and even on this basis, various tools and questionnaires focused on each specific human characteristic have been developed in the human-like robots' field (Andrighetto et al., 2025; Ruijten et al., 2019). In previous studies, the various effects of this characteristic of robots have been discussed including both positive and negative aspects.

#### *2.1.1. pros*

One of the pros of anthropomorphizing is smoothness. Smoothness in an interaction between a human and a robot means that the robot can provide "without jerk" movement and trajectory profile (Erol & Sarkar, 2007). In recent studies, anthropomorphizing robots are tried to provide a smooth human-robot interactions (Fasola & Mataric, 2012). So, in this case, robots with more human-likeness features perceived smoother. Another positive aspect is increasing the chance of comprehension and social companionship. Unlike the previous one, which was related to the physical characteristics of the robot, such as the way its arms move, and can be of interest to rehabilitation specialists, in this second case, the benefits of using the human-like robots in social parameters have been discussed (Kühne et al., 2024). For example, sometimes this mode increases the learnability of students from a robot teacher or even the increase of human trust in the robot at the start of a collaboration is mentioned (Roesler et al., 2024).

#### *2.1.2. cons*

One of the concepts that is mainly referred to as the design flaw of an anthropomorphized robot is a case called "uncanny valley" (Yam et al., 2021). This theory is hypothesized based on human emotional response to the degree of anthropomorphism of a robot (Di Natale et al., 2023). The uncanny valley hypothesis predicts that an entity appearing almost human will risk

eliciting unpleasant feelings in human partners (Appel et al., 2020). Also, one of the most important criticisms of humanizing robots is related to hiding their reality. Some researchers believe that by over-humanizing robots, their robotic beings (being a machines) should not be forgotten. Because companionship, cooperation and even living with a machine has its own requirements, which may be different in human-human communication (Tan et al., 2020). They believed that it can leads to excessive attachment, social isolation, fear, and violation of privacy (Vaitonytė et al., 2023).

*2.1.3. Crucial summation*

The list of advantages and disadvantages of using anthropomorphism in robots is not limited to these cases but is beyond the aim of this systematic literature review. But these important ones are given to remind the importance of this concept in human interaction with technology. It is always important to consider the balance between anthropomorphism and the functional characteristics of the robot to have an optimum user experience. Because by using this feature inappropriately, the mentioned advantages and disadvantages may even change their places, and in fact, when a greater advantage is expected, more defects in the interaction could happen.

### *2.2. Some theoretical supports for adjusting trust to anthropomorphized AVs*

Some theories emphasize how giving human-like qualities to technology can increase trust. "*Social Presence Theory*" (Short et al., 1976) suggests that the more a non-human agent can evoke the feeling of a real person, the richer the user's interaction and, consequently, their emotional connection and trust (Lu et al., 2016). Similarly, the "*Computers Are Social Actors*" theory (Nass et al., 1994) implies that humans unconsciously apply social rules and norms, typically reserved for interpersonal interactions, to computers and technology. This means that when a device exhibits human-like traits, people tend to treat it as a social entity, which includes extending trust (Gambino et al., 2020; Kim et al., 2025). This one is also extended as a new theory applied for cars entitled "*Cars As Social Agents*" (Babel et al., 2023). Furthermore, while primarily an interpersonal concept, "*Attachment Theory*" (Mitchell & Jeon, 2025) has been applied to human-technology relationships, positing that if anthropomorphized technology can create a sense of closeness and security, it can foster a deeper bond and, by extension, trust (Nafees & Sujood, 2024). Finally, the general impact of "*Social Cues*" (Gutuleac et al., 2024) in social psychology highlights that human-like signals (e.g., eye contact, smiles, tone of voice) in any interaction increase intimacy and understanding. When these cues are incorporated into technology, they activate the same psychological mechanisms that lead to greater trust (Xu & Ma, 2025).

### *2.3. From anthropomorphized robots towards anthropomorphized AVs: A rationale for a new systematic review and meta-analysis*

The closest domain related to anthropomorphic interfaces in the context of AVs is social robots. In this field, comprehensive studies, including systematic reviews, have been written to summarize the role of these interfaces in building trust (and, to a lesser extent, diminishing trust in some studies) in robots (Damholdt et al., 2023; Dubois-Sage et al., 2023; Roesler et al., 2021). However, inherent differences between human-robot and human-AV interactions are not ignorable (Zhang et al., 2022). For instance, human-robot interaction (HRI) typically occurs in a stationary environment, while dynamic interactions with AV happen in a moving space. This movement, along with the inherent risks associated with driving, may influence the relationship between anthropomorphic interfaces and trust in AV. Additionally, interactions with humanoid robots usually take place in larger and not confined spaces (Alireza Mortezapour, 2025), such as shopping malls, hotels, hospitals, and homes, while most interactions with AVs are assumed to be confined to the small space inside the vehicle's cabin. This distinction seems to have an impact on the anthropomorphism-trust relationship through altering the interaction pattern and reshaping expectations of the system (theory of mind concept (Shank et al., 2019)). From another perspective, in interactions with robots, users typically direct much of their attention (or a significant portion of it) toward the robot. At least in lower degree of automated driving (L3), due to some necessitated situation awareness for potential take-overs (Mosaferchi et al., 2025; Shi et al., 2024), this interaction is different, which could significantly affect the trust.

## 3. Methods

This systematic review and meta-analysis was conducted in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-analysis) guidelines (Moher et al., 2010) to investigate the effect of anthropomorphism on human trust in AVs. The protocol was registered on the Open Science Framework on 2nd of April 2024, and can be accessed at https://doi.org/10.17605/OSF.IO/7P9NZ.

1.1.Search strategy

The original search was done in December 2023, and a follow-up search was conducted in February 2025, along with forward and backward citation searches. The search period for the current study was set from January 1, 1990 to February 10, 2025. While a study recommends searching at least two databases to reduce the risk of missing eligible studies (Ewald et al.,

2022), another suggests that conducting systematic reviews with two databases can cause the risk of publication bias (Mathew, 2024). In addition, the significance of including grey literature, such as conference papers and dissertations, to decrease the risk of publication bias has been emphasised (Chandler et al., 2019). Therefore, a search through 7 databases including Scopus, PsycINFO, Web of Science, ProQuest, Google Scholar, IEEE, and ACM, was conducted to reduce the risk of missing eligible studies. The selection of databases was guided by the research focus, which emphasizes psychological research and human-computer interaction. A combination of terms was used to search in databases (see Table 1). According to the guideline by (Keele, 2007), authors first conducted an initial search to scan titles, abstracts, and keywords of core articles and relevant systematic reviews to extract keywords. A consultation with experts in the field was performed to finalize the search terms. Supplementary appendix A provides the search syntax used for Scopus, which was modified to align with the syntax requirements of other databases.

Table 1. Search terms.

| **Anthropomorphism Keywords (OR)** | **AND** | **Vehicle Automation Keywords (OR)** | **AND** | **Trust** |
|---|---|---|---|---|
| "anthropomorph*", "humanoid", "human likeness", "human-likeness", "humanness", "man likeness", "humanlike", "human-like", "humaneness", "person like", "person-like", "caricature", "speech", "speech-based", "conversat*", "vocal", "conversational agent", "humanis*", "humaniz*", "personif*", "anima*", "audi*", "voice", "cartoon*" | | "automated car*", "autonomous car*", "selfdriving", "self-driving", "self driving", "driverless", "vehicle automation", "automated driving", "autonomous vehicle*", "automated vehicle*" | | "trust" |

1.2.Inclusion and exclusion criteria

Only studies examining the interaction between humans and driving automation level 3 or higher, as defined by the SAE taxonomy (SAE, 2021), were included. Eligible studies were those that included the incorporation of human participants who either viewed or participated directly in interaction with AVs. Surveys utilizing questionnaires, without the involvement of human subjects and automated driving systems (e.g., (Wu et al., 2023b)), were excluded from the study. This exclusion ensures that the evaluation of trust factors is based on real-life experiences rather than imagination, aligning with the concept of dynamic trust. Studies had to

examine varying levels of anthropomorphism as an independent variable and report human trust in interactions with AVs. To ensure the comprehensiveness of the results and minimize the risk of publication bias, all types of publications were eligible, including conference papers, journal articles, and dissertations (Rothstein et al., 2005). Finally, studies had to be written in English.

### 1.3.Study selection and data extraction

A total of 3008 papers were extracted from seven databases and imported to the EndNote (version 21) software to remove duplicates. To determine whether the remaining articles (2177) meet the inclusion criteria, two authors (AM, SM) independently screened the titles and abstracts, resulting in 237 articles. Then, the full texts of the remaining articles were screened for eligibility (AM, SM) and disagreements were resolved in consultation with a third reviewer (AA). Authors treated multiple experiments within each study as separate samples. The PRISMA diagram (Fig. 1) shows the study selection process and reasons for excluding studies. A data collection form was created in an Excel spreadsheet (see supplementary appendix B) and finalized after a trial data extraction process conducted by two authors (AM, AA) on several research papers. The form was used to summarize the following information from each article: (i) bibliographic details (title of the study, authors, publication year, country), (ii) methodology (study design, theoretical framework), (iii) personal characteristics (e.g., age, gender), (iv) details of the experiment (participant sample size, characteristics of the driving scenario, SAE automation level, HMI modality, level of anthropomorphism, role of participants, driving simulation, and presence of non-driving related task (NDRT).

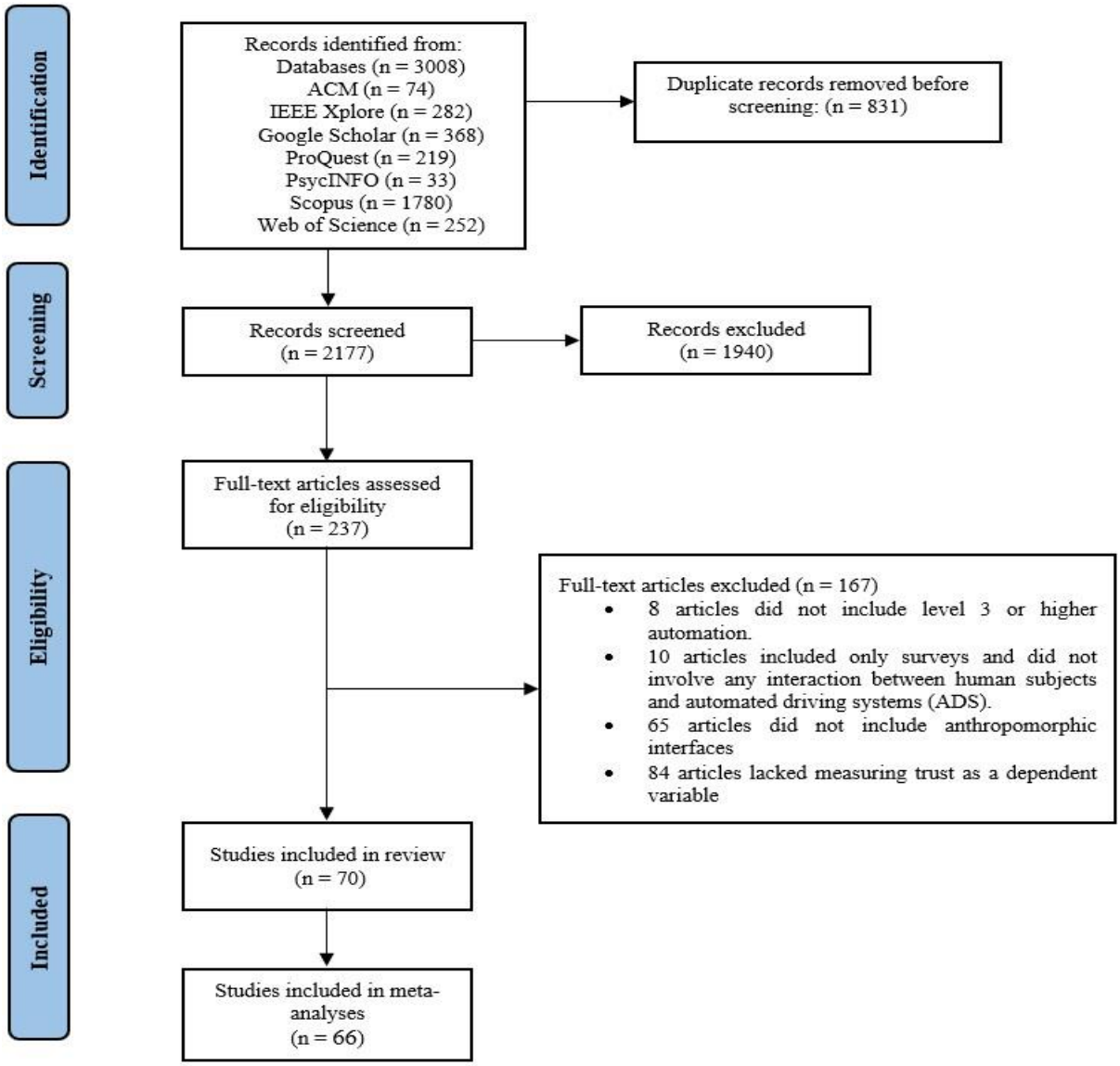



*Fig. 1.* Systematic literature review process based on PRISMA method.

1.4.Risk of bias assessment

The quality of included studies was assessed using Joanna Briggs Institute (JBI) critical appraisal tool for analytical cross-sectional studies (Moola et al., 2017). This is a validated and widely used tool for appraising the quality of cross-sectional research, which includes 8 questions related to methodological criteria: Q1 (Were the criteria for inclusion in the sample clearly defined?), Q2 (Were the study subjects and the setting described in detail?), Q3 (Was the exposure measured in a valid and reliable way?), Q4 (Were objective, standard criteria used for measurement of the condition?), Q5 (Were confounding factors identified?), Q6 (Were strategies to deal with confounding factors stated?), Q7 (Were the outcomes measured in a valid and reliable way?), and Q8 (Was appropriate statistical analysis used?). As Q4 assesses whether studies provide evidence that specified diagnostic methods or definitions were used for

including patients when they were selected based on a particular diagnosis or definition (Moola et al., 2017). As the included studies did not require a specific diagnosis or definition for participation, this question was not applicable to the studies in the current systematic review.

The questions should be answered according to: Yes, No, Unclear, or Not/Applicable. Each question answered with "yes" was assigned 1 point, while "no" and "unclear" received 0 points. Responses marked as "not applicable" were excluded from the scoring. In addition to each study's total score, the relevant percentage was also reported to indicate its quality level. For instance, a study with 7 ‘yes’ responses out of 7 questions received a total score of 7, corresponding to 100%. Studies with a JBI score above 75% were classified as high quality, those scoring between 50% and 75% were considered moderate quality, and studies with a JBI score below 50% were categorized as low quality (Alluhaybi et al., 2023). Two researchers (AM, SM) independently rated the quality of each study using the checklist and disagreements in ratings were resolved in consultation with a third reviewer (AA).

### 1.5.Data synthesis

From each study, sample sizes, means, and standard deviations (SDs) of the trust scores in AVs between different levels of anthropomorphism were extracted. If means and SDs were reported in graphs and charts, Web Plot Digitizer application (https://automeris.io/) was used to extract the values. If extracting means and SDs through the methods mentioned above was not possible, mean and SD values was calculated through the methods introduced by (Wan et al., 2014). Additionally, SD values were generated by converting standard errors (Sestino & D'Angelo) or confidence intervals (CIs), using formulas provided in the Cochrane Handbook (section 6.5.2.2) (Chandler et al., 2019). Some eligible studies did not provide sufficient data to calculate an effect size. In such cases, the authors were contacted up to three times at one-week intervals to request the raw data necessary for effect size calculation. Studies were excluded from meta-analyses if authors did not respond to requests for further information. The standardized mean differences (SMD) rather than weighted mean differences (WMD) was calculated for each study because, although all studies measured trust in AVs at varying levels of anthropomorphism, they used different questionnaires that featured different rating scores and items (Andrade, 2020). Hedge's g was used to calculate SMD as it is particularly suitable when the sample sizes are small (<20) (Cumming, 2013). To interpret the strength of the effect, the following SMD cut-offs were used: Trivial (< 0.20), small (0.20–0.49), moderate (0.50–

0.79) and large (≥ 0.80) (Cohen, 2013). To account for expected heterogeneity across studies, all analyses were performed using the random-effects model (Chandler et al., 2019).

As shown in Table *2* each individual study investigated how varying levels of anthropomorphism influence human trust in AVs. Specifically, they examined one or more of the following comparisons: (1) anthropomorphic versus non-anthropomorphic interfaces; (2) superficial versus deep anthropomorphism; and (3) anthropomorphic interface versus the lack of interface. Based on these comparisons, authors conducted three separate meta-analyses to investigate the impact of different level of anthropomorphism on human trust in AVs. Heterogeneity across studies was assessed using the $I^2$ statistic. Low, moderate, and high levels of heterogeneity were defined by $I^2$ outcomes of 25%, 50%, and 75%, respectively (Borenstein et al., 2021). In the case of high and significant heterogeneity, subgroup analyses and meta-regressions were conducted to identify the sources of heterogeneity, and the potential effects on the results are discussed. Given that the primary aim of our study was to compare anthropomorphic and non-anthropomorphic interfaces, subgroup analyses and meta-regressions were performed exclusively within this meta-analysis. Subgroup analyses were limited to: (1) anthropomorphic interface modality (auditory-speech, visual, and visual_auditory-speech) because anthropomorphic characteristic is presented in different modalities. Some researchers see anthropomorphism as a human-like visual cue (e.g. a human eye or a humanized smile) and others as a human-like voice (specially empowered by Large Language Models (LLMs)) which is integrated to a technology such as a AV (Mahadevan et al., 2018; Shin et al., 2025); (2) level of automation (level 3 vs. level 5), because it has been shown that level of automation of the vehicles is effective in perceived anthropomorphism (Messer et al., 2024); (3) the quality of studies (low, moderate, and high), (4) the presence of an NDRT (yes vs. no) (5) the role of participants (driver, and pedestrian) (Zhou et al., 2025).; and (6) the driving simulation (driving simulator, real car, virtual reality, and video-picture) because they cannot produce completely identical impact and are not comparable to each other in the Human-AV interaction (Fuest et al., 2020).

Also participants' views of technology anthropomorphism may differ based on their individual characteristics, such as age (Letheren et al., 2016) and gender (Yu & Ngan, 2019). Meta-regression was therefore performed for continuous variables of age of participants and female percentage. For all results, a *p*-value of 0.05 was considered as significant, and all analyses were performed using the RStudio version 2025.05.1 (Team, 2020) and the "meta" package (Balduzzi et al., 2019).

Publication bias was assessed with visual inspection and statistical test using the funnel plot and Egger bias test, if ≥10 studies were available for a single meta-analysis (Egger et al., 1997). A $p < 0.05$ based on the Egger's test along with an asymmetry in the funnel plot are considered as indications of publication bias. To enhance the robustness of findings, authors conducted a sensitivity analysis by sequentially excluding one study from the analysis (i.e., leave-one-out approach). Data and analysis code are available in the following link (https://github.com/ali-arabian/anthropomorphism_trust_AVs).

# 2. Results

## 2.1. Study characteristics

As shown in Table *2*, 70 studies were included in the analysis, of which 34 studies (56.66%) compared the impact of anthropomorphic versus non-anthropomorphic interfaces on human trust in AVs [study IDs according to Table 2: 1-3, 5-8, 10-13, 16, 23-26, 29, 31-32, 35-36, 38-42, 45, 47-48, 51, 60, 64, 68-69], 50 studies compared superficial versus deep anthropomorphism [study IDs: 1, 3-4, 6, 9-10, 14-15, 17-22, 26, 29-34, 36-38, 43-46, 48-50, 52-70], and 13 studies compared anthropomorphic interface versus the lack of interface (I.e., baseline) [study IDs: 8-9, 15, 18, 27-28, 37, 45, 49, 53, 56-58]. The mean sample size per study was 84.3 (SD = 119.9), ranging from 8 to 731 participants, with an overall total of 5,903. These eligible studies were published between 2015 and 2025, representing three continents: Europe, Asia, and America. The mean age of participants across studies was 29.01 years (SD = 8.09), ranged between 10.73 and 60.07 years, and the mean proportion of female participants across studies was 45.58% (SD = 12.7%), ranged from a minimum of 9% to a maximum of 67%. Regarding the engagement with NDRT, only 16 studies included it, while the remaining 54 studies did not. Of the included studies, 30 examined human interaction with L3 driving automation, 2 with L4, and the remaining 38 focused on fully AVs (L5). In 54 out of 70 studies, participants assumed the role of drivers, while in 10 studies they acted as pedestrians and in 6 studies as passengers. Regarding driving simulation, the studies were classified into three categories: (i) *Simulator* – 37 studies recreated the driving environment using a driving simulator or pedestrian simulation lab; (ii) *Video/Picture* – 19 studies used videos or images to represent the driving environment; (iii) *Virtual Reality (Vrkljan & Anaby)* – 8 studies used VR headset to represent the driving environment; and (iv) *Real Car* – 6 studies conducted in real-world driving conditions.

2.2.Risk of bias assessment of the included studies

Table 2 shows the classification of the methodological quality for each included study. 48.57% of included studies (34/70) were classified as high quality, while 30 out of 70 studies were of moderate quality (42.86%). The remaining studies (6/70) were of low quality (8.57%). Common quality assessment criteria that were not met by studies included identifying confounding factors and strategies to deal with them, clearly defining inclusion criteria, using appropriate statistical analysis, and measuring outcomes in a valid and reliable way, respectively. See supplementary appendix C for further details of the quality assessment.

Table 2. Characteristics of included studies.

| Study ID | Study | N | Age: Year (SD); Min/Max | Gender: F/M | Level of automation | Modality | | | | Role of participants | Driving simulation | NDRT | JBI quality criteria score |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Deep Anthro | Superficial Anthro | Comparison | Baseline (Without modality) | | | | |
| **1** | (Aleva et al., 2024) | 28 | 27.2(9.2); 19/59 | 5/23 | L5 | Visual | Visual | Visual | No | Pedestrian | Virtual Reality | No | 57% Moderate |
| **2** | (Antrobus et al., 2018) | 23 | 28.7(8.69) | 4/19 | L5 | Auditory-Speech | - | Visual | No | Passenger | Simulator | No | 57% Moderate |
| **3** | (Aremyr et al., 2019) | 11 | 47(-); 25/68 | 5/6 | L5 | Visual Auditory-Speech | Visual Auditory-Speech | Visual Auditory | No | Passenger | Real Car | No | 100% High |
| **4** | (Atchley et al., 2022) | 92 | 21.3 (5.7) | | L5 | Visual | Visual | - | No | Driver | Video-Picture | No | 43% Low |
| **5** | (Bonneviot et al., 2023) | 731 | 41.58(12.70); 18/65 | 383/348 | L5 | Visual Auditory-Speech | - | Visual Auditory | No | Pedestrian | Video-Picture | No | 71% Moderate |
| **6** | (Choi et al., 2022) | 45 | Median = 23(-); 18/34 / | 29/16 | L5 | Visual | Visual | Visual | No | Pedestrian | Virtual Reality | No | 86% High |
| **7** | (Chung et al., 2024) | 38 | 10.73(1.12); 10/12 | 13/25 | L5 | Visual | - | Visual | No | Pedestrian | Video-Picture | No | 86% High |
| **8** | (Colley et al., 2022) | 226 | 31.04(9.71) | 94/129/2 n/a | L5 | Auditory-Speech | - | Visual | Yes | Driver | Video-Picture | No | 100% High |
| **9** | (Du et al., 2019) | 32 | 26.9(6.3) | 11/21 | L3 | Auditory-Speech | Auditory-Speech | - | Yes | Driver | Simulator | No | 71% Moderate |
| **10** | (Faas et al., 2020) | 59 | 42.98(14.85); 18/66 | 27/32 | L5 | Visual | Visual | Visual | No | Pedestrian | Real Car | No | 86% High |
| **11** | (Flohr et al., 2021) | 34 | 29.9(9.8); 19/61 | 15/18/1 n/a | L5 | Visual | - | Visual | No | Passenger | Video-Picture | No | 86% High |
| **12** | (Forster et al., 2017) | 17 | 29.0(8.12); 22/56 | 7/10 | L3 | Auditory-Speech | - | Visual | No | Driver | Simulator | Yes | 71% Moderate |
| **13** | (Frison et al., 2019) | 18 | 25.11(10.35); 19/65 | 12/6 | L3 | Auditory-Speech | - | Visual | No | Driver | Simulator | Yes | 57% Moderate |
| **14** | (He & Burns, 2022) | 68 | 34.6(-); 20/62 | 23/44 | L5 | Auditory-Speech | Auditory-Speech | - | No | Driver | Video-Picture | No | 57% Moderate |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **15** | (Hock et al., 2016) | 38 | 24(3.54) | 25/13 | L3 | Visual Auditory-Speech | Auditory-Speech | - | Yes | Driver | Virtual Reality | No | 71% Moderate |
| **16** | (Hong et al., 2025) | 70 | 30.0(10.18); 19-78 | 33/37 | L5 | Auditory-Speech | - | Visual | No | Driver | Simulator | Yes | 43% Low |
| **17** | (Ji et al., 2019) | 30 | 24.6(2.58); 21/35 | 15/15 | L3 | Auditory-Speech | Auditory-Speech | - | No | Driver | Simulator | No | 86% High |
| **18** | (Kaufman et al., 2024) | 41 | _ | _ | L5 | Auditory-Speech | Auditory-Speech | - | Yes | Driver | Simulator | No | 86% High |
| **19** | (Kettle et al., 2023) | 499 | 26.04(10.23); 18/74 | 277/211 | L3 | Visual | Visual | - | No | Driver | Video-Picture | No | 100% High |
| **20** | (Y. M. Kim et al., 2024) | 30 | 23.10(1.68); 20/28 | 17/13 | L5 | Auditory-Speech | Auditory-Speech | - | No | Driver | Simulator | No | 86% High |
| **21** | (Kim et al., 2025) | 84 | 24.7(3.33); 19/32 | 51/32 | L3 | Auditory-Speech | Auditory-Speech | - | No | Driver | Simulator | No | 100% High |
| **22** | (Knutzen et al., 2019) | 56 | 24.78(4.89); 19/52 | 5/51 | L3 | Visual | Visual | - | No | Driver | Simulator | No | 86% High |
| **23** | (Ko et al., 2022) | 21 | 19(0.9) | 4/19 | L3 | Auditory-Speech | - | Auditory | No | Driver | Simulator | Yes | 86% High |
| **24** | (Kohn et al., 2018) | 52 | 37(12) | 30/22 | L5 | Visual | - | Visual | No | Driver | Video-Picture | No | 71% Moderate |
| **25** | (Körber, Prasch, et al., 2018) | 40 | 25.2(2.60); 21/30 | 20/20 | L3 | Auditory-Speech | - | Visual | No | Driver | Simulator | Yes | 100% High |
| **26** | (Kraus et al., 2016) | 16 | 27.63(8.96) | _ | L3 | Visual Auditory-Speech | Visual Auditory-Speech | Auditory | No | Driver | Simulator | No | 71% Moderate |
| **27** | (Kühnlenz & Kühnlenz, 2023) | 189 | 27.5(12.7); 18/75 | 84/104 | L5 | Auditory-Speech | - | - | Yes | Driver | Video-Picture | No | 57% Moderate |
| **28** | (Kunze et al., 2019) | 34 | 30.8(9.5) | 14/20 | L3 | Visual | - | - | Yes | Driver | Simulator | Yes | 71% Moderate |
| **29** | (Large et al., 2019) | 34 | 40(11.4); 21/58 | 17/17 | L5 | Auditory-Speech | Auditory-Speech | Visual | No | Driver | Real Car | No | 86% High |
| **30** | (Large et al., 2025) | 41 | _ | _ | L5 | Visual | Visual | - | No | Pedestrian | Real Car | No | 14% Low |
| **31** | (Lawson-Guidigbe, | 146 | 36.92(14.04) | 88/58 | L5 | Visual | Visual | Visual | No | Driver | Video-Picture | No | 57% Moderate |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **32** | Louveton, et al., 2023) (Lawson-Guidigbe, Amokrane-Ferka, et al., 2023b) | 36 | _ | _ | L3 | Visual Auditory-Speech | Auditory-Speech | Visual Auditory Tactile | No | Driver | Simulator | Yes | 71% Moderate |
| **33** | (Lee & Lee, 2022) | 191 | 41.16(11.12); 24/74 | 75/116 | L3 | Auditory-Speech | Auditory-Speech | - | No | Driver | Video-Picture | No | 71% Moderate |
| **34** | (Lee & Lee, 2022) | 96 | 23.84(3); 18/34 | 48/48 | L3 | Auditory-Speech | Auditory-Speech | - | No | Driver | Simulator | No | 57% Moderate |
| **35** | (Lee et al., 2015) | 89 | 23.65(2.21); 19/29 | 45/44 | L5 | Visual | - | Visual | No | Pedestrian | Simulator | No | 86% High |
| **36** | (Lin, 2024) | 314 | 26(-); 21/73 | 183/127 | L5 | Visual | Visual | Visual | No | Driver | Video-Picture | No | 57% Moderate |
| **37** | (Liu et al., 2024) | 257 | 19.63(1.91); 17/33 | 97/160 | L3 | Auditory-Speech | Auditory-Speech | - | Yes | Passenger | Real Car | No | 86% High |
| **38** | (Löcken et al., 2019) | 20 | 27(7.2) | 7/13 | L5 | Visual | Visual | Visual | No | Pedestrian | Virtual Reality | No | 57% Moderate |
| **39** | (Ma et al., 2020) | 8 | -(-); 20/30 | 4/4 | L5 | Visual Auditory-Speech | - | Visual | No | Driver | Video-Picture | No | 71% Moderate |
| **40** | (Meinhardt et al., 2024) | 14 | 60.07(12.46) | 7/7 | L5 | Auditory-Speech | - | Tactile | No | Driver | Simulator | No | 100% High |
| **41** | (Nadri et al., 2024) | 20 | 24.3(6.32) | 6/13/1n/a | L4 | Visual Auditory-Speech | - | Visual Auditory | No | Driver | Simulator | No | 100% High |
| **42** | (Niu et al., 2018) | 39 | 27.21(7.22) | 15/24 | L5 | Visual | - | Visual | No | Driver | Simulator | No | 100% High |
| **43** | (Park et al., 2024a) | 30 | 23.1(1.8) | 17/13 | L5 | Auditory-Speech | Auditory-Speech | - | No | Driver | Simulator | No | 100% High |
| **44** | (Pavone & Desveaud, 2025) | 309 | 43.14(12.93); 20/80 | 152/157 | L5 | Auditory-Speech | Auditory-Speech | - | No | Driver | Video-Picture | No | 86% High |
| **45** | (Pratticò et al., 2025) | 24 | 27.5(7.1); 21/58 | 9/15 | L3 | Visual Auditory-Speech | Auditory-Speech | Visual Auditory | Yes | Driver | Virtual Reality | Yes | 57% Moderate |
| **46** | (Premstaller et al., 2023) | 52 | 29.58(12.60); 20/61 | 26/23/3n/a | L3 | Visual | Visual | - | No | Driver | Video-Picture | No | 57% Moderate |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **47** | (Ruijten et al., 2018) | 57 | 24.5(1.96); 20/30 | 13/44 | L3 | Visual Auditory-Speech | - | Visual | No | Driver | Simulator | No | 57% Moderate |
| **48** | (Schneider et al., 2017) | 32 | 23.31(3.15); 18/30 | 13/19 | L3 | Visual | Visual | Visual | No | Driver | Simulator | Yes | 57% Moderate |
| **49** | (Schuß et al., 2024) | 136 | 32.51(16.17); 18/78 | 68/76 | L5 | Visual Auditory-Speech | Visual Auditory-Speech | - | Yes | Passenger | Video-Picture | No | 57% Moderate |
| **50** | (Shin et al., 2025) | 89 | 29.8(8.7); 20/62 | 52/37 | L5 | Auditory-Speech | Auditory-Speech | - | No | Driver | Video-Picture | No | 86% High |
| **51** | (Song et al., 2023) | 60 | - | - | L3 | Auditory-Speech | - | Visual | No | Driver | Simulator | No | 43% Low |
| **52** | (Stampf et al., 2024) | 14 | 25.07(3.1); 21/34 | 6/8 | L5 | Auditory-Speech | Auditory-Speech | - | No | Driver | Virtual Reality | No | 43% Low |
| **53** | (Sun et al., 2024) | 55 | 23.9(3.9); 19/44 | 32/23 | L4 | Auditory-Speech | Auditory-Speech | - | Yes | Driver | Virtual Reality | Yes | 86% High |
| **54** | (Torggler et al., 2022) | 25 | 23.28(2.74) | 8/17 | L3 | Auditory-Speech | Auditory-Speech | - | No | Driver | Simulator | Yes | 57% Moderate |
| **55** | (Verberne et al., 2015) | 111 | 39.8(12.7) | 58/53 | L5 | Visual | Visual | - | No | Passenger | Simulator | No | 100% High |
| **56** | (Wallbridge et al., 2024) | 296 | 40.33(13.72); 18/73 | 157/135 | L5 | Visual Auditory-Speech | Visual | - | Yes | Driver | Video-Picture | No | 100% High |
| **57** | (Wallbridge et al., 2024) | 199 | 42.05(13.09); 18/80 | 91/108 | L5 | Visual Auditory-Speech | Visual | - | Yes | Driver | Video-Picture | No | 100% High |
| **58** | (Wallbridge et al., 2024) | 192 | 41.52(12.48); 18/80 | 110/80 | L5 | Visual Auditory-Speech | Visual | - | Yes | Driver | Video-Picture | No | 100% High |
| **59** | (M. Wang et al., 2021) | 24 | 23.13(4.59) | 8/16 | L5 | Visual Auditory-Speech | Auditory-Speech | - | No | Driver | Simulator | No | 57% Moderate |
| **60** | (P. Wang et al., 2021) | 10 | 26.36(4.06) | 6/4 | L5 | Visual | Visual | Visual | No | Pedestrian | Real Car | No | 71% Moderate |
| **61** | (Wang et al., 2022) | 24 | 23.12(4.49); 19/33 | 7/17 | L3 | Visual Auditory-Speech | Auditory-Speech | - | No | Driver | Simulator | No | 57% Moderate |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **62** | (Wang et al., 2024) | 24 | 23.12(4.49); 19/33 | 7/14 | L3 | Visual Auditory-Speech | Auditory-Speech | - | No | Driver | Simulator | No | 71% Moderate |
| **63** | (Wang et al., 2024) | 24 | 23.13(4.59); 18/36 | 8/13 | L5 | Visual Auditory-Speech | Auditory-Speech | - | No | Driver | Simulator | No | 71% Moderate |
| **64** | (Wiehr et al., 2021) | 23 | 25.13(2.48); 21/29 | 15/8 | L3 | Auditory-Speech | Auditory-Speech | Auditory | No | Driver | Simulator | Yes | 86% High |
| **65** | (Wong et al., 2019) | 20 | 26.3(7.24); 21/48 | 8/12 | L3 | Auditory-Speech | Auditory-Speech | - | No | Driver | Simulator | Yes | 43% Low |
| **66** | (Yin et al., 2024) | 25 | 26.3(3.1); 20/29 | 12/13 | L3 | Auditory-Speech | Auditory-Speech | - | No | Driver | Simulator | Yes | 100% High |
| **67** | (Yoo et al., 2022) | 41 | 27.83(8.47) | 22/19 | L3 | Auditory-Speech | Auditory-Speech | - | No | Driver | Simulator | Yes | 86% High |
| **68** | (Y. Zhang et al., 2023) | 30 | 26.03(2.68); 23/33 | 14/16 | L3 | Auditory-Speech | Auditory-Speech | Auditory | No | Driver | Simulator | No | 100% High |
| **69** | (Zhang et al., 2024) | 10 | -(-); 23/31 | 6/4 | L5 | Auditory-Speech | Auditory-Speech | Visual | No | Pedestrian | Virtual Reality | No | 86% High |
| **70** | (Zhao et al., 2024) | 40 | 23.44(2.97); 20/31 | 20/20 | L3 | Auditory-Speech | Auditory-Speech | Visual | No | Driver | Simulator | Yes | 100% High |

2.3.Meta-analyses

Data pooling from 34 studies indicated no significant difference in human trust in AVs when interacting with anthropomorphic compared to non-anthropomorphic interfaces (SMD = 0.04, 95% CI [-0.25, 0.33], $p$ = 0.797, trivial effect, Fig 2). The between-study heterogeneity was high and significant ($I^2$ = 82%, $p$ < 0.001). Similarly, the level of anthropomorphism (superficial versus deep) (k = 50; SMD = 0.05, 95% CI [-0.14, 0.23], $p$ = 0.623; $I^2$ = 83%, $p$ < 0.001, Fig. 3) and the presence of anthropomorphic interface versus baseline, i.e., the lack of interface (k = 13; SMD = 0.15, 95% CI [-0.02, 0.33], $p$ = 0.091; $I^2$ = 43%, $p$ = 0.051, Fig. 4) showed non-significant trivial effect on human trust in AVs.

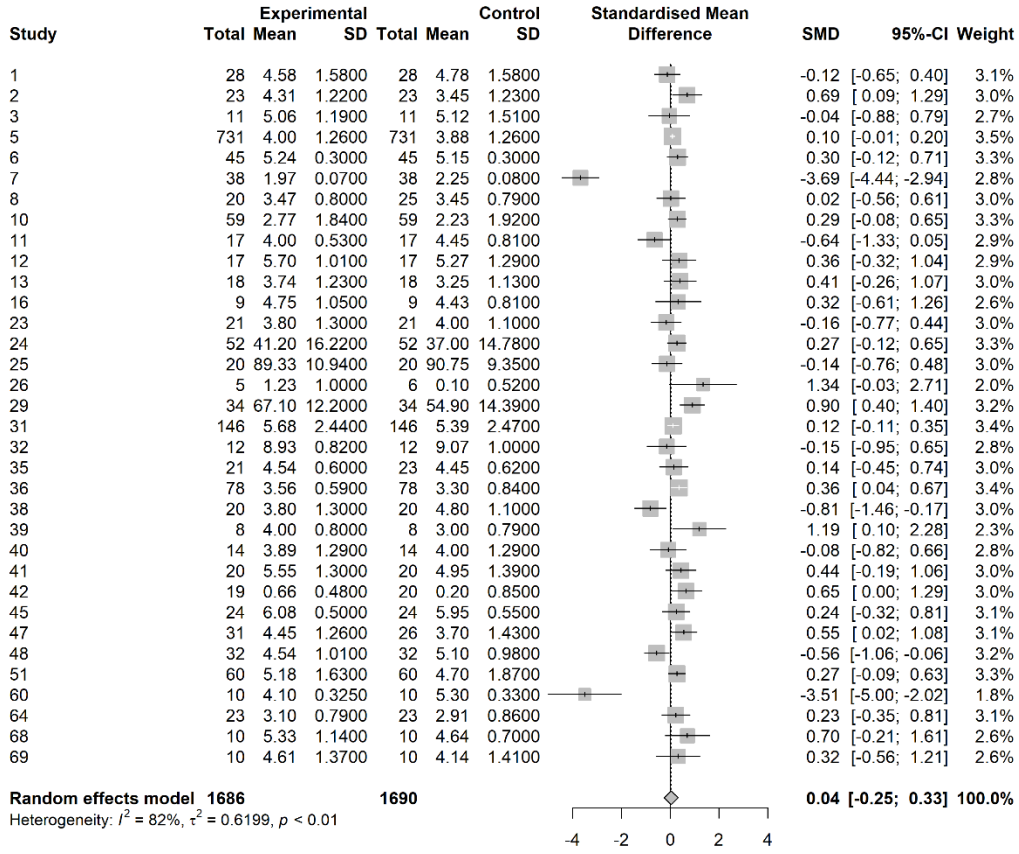



**Fig 2.** Forest plot of the effect of anthropomorphic interface on human trust in AVs compared to non-anthropomorphic interface.

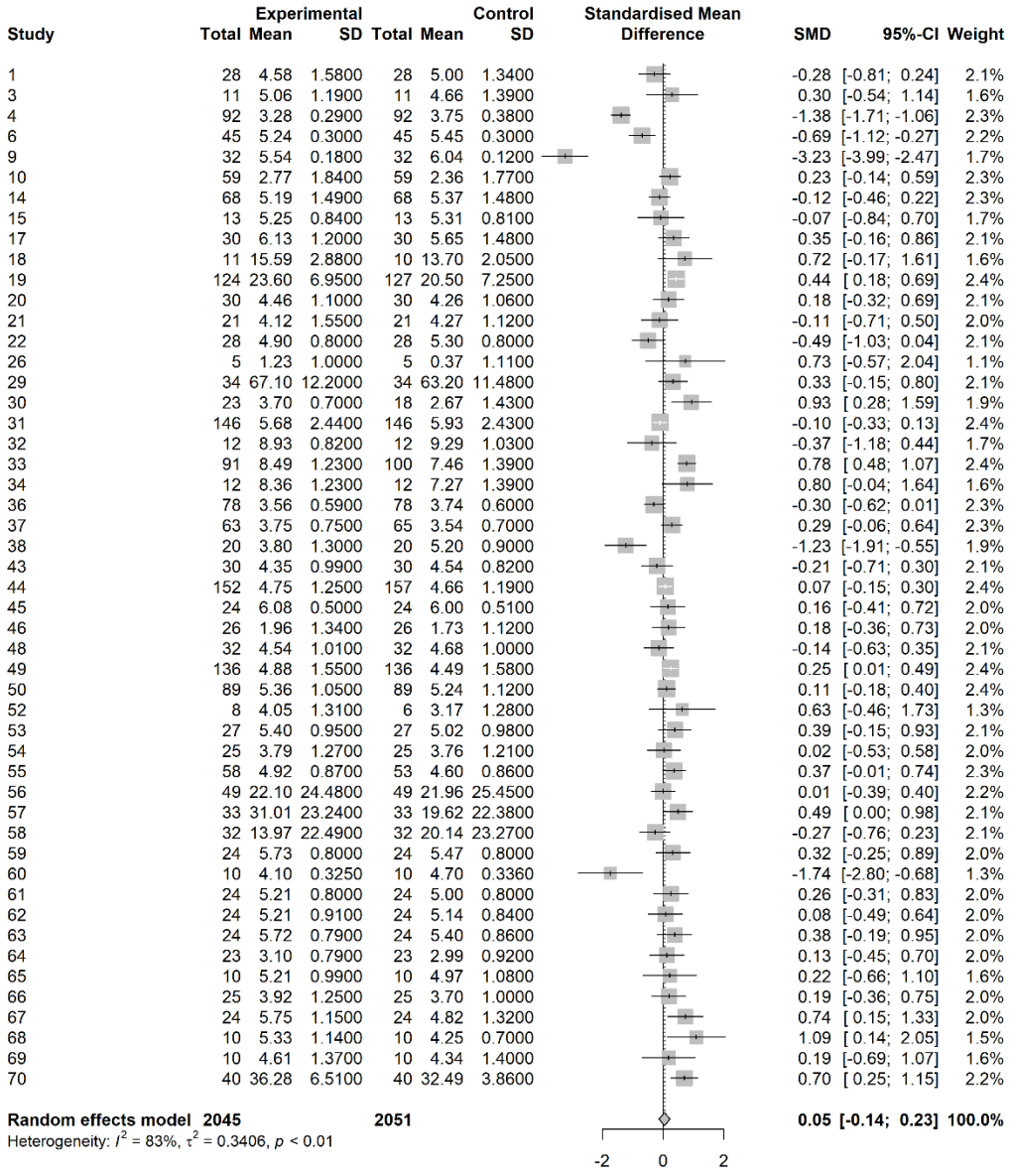



**Fig. 3.** Forest plot of the effect of deep anthropomorphism on human trust in AVs compared to superficial anthropomorphism.

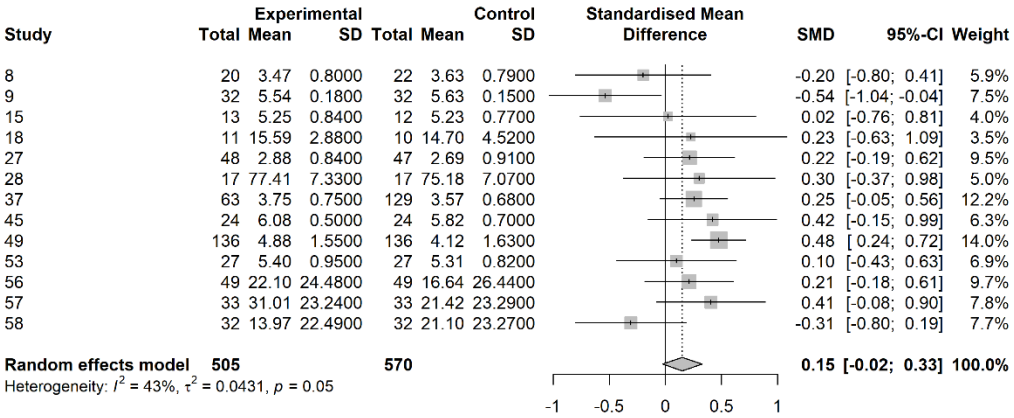



**Fig. 4.** Forest plot of the effect of anthropomorphic interface on human trust in AVs compared to baseline (i.e., the lack of interface).

2.4.Heterogeneity: subgroup analyses and meta-regression

Due to the high heterogeneity across studies comparing anthropomorphic and non-anthropomorphic interfaces ($I^2 = 82\%$, $p < 0.001$), subgroup and meta-regression analyses were conducted to identify potential moderators influencing the impact of anthropomorphism on human trust in AVs, as highlighted in the relevant literature. The results of the subgroup analyses are shown in Table 3. In addition, meta regression was computed for two continuous variables (i.e., participants' average age and female percentage). The results showed a marginal significant predictive value of age on the impact of anthropomorphism on human trust in AVs ($p = 0.07$); each year increase in average age was associated with a 0.031 increase in the SMD of trust (Table 4). Results of meta-regressions performed on female percentage showed no significant effect on the impact of anthropomorphism on human trust in AVs. See supplementary appendix D for graphical representation of subgroup analyses and meta regressions.

Table 3. Subgroup meta-analysis results

| Subgroups | K | Between group heterogeneity | SMD [CI 95 %] | $I^2$ | Conclusion |
|---|---|---|---|---|---|
| **Overall** | 34 | | 0.04 [-0.25; 0.33] | 82% | |
| **Anthropomorphism modality** | | | | | The presence of a speech-based anthropomorphic interface, either on its own or combined with a visual interface, had a significantly small positive effect on human trust in automated vehicles. In contrast, the unimodal visual anthropomorphic interface did not lead to a statistically significant effect. The auditory and audiovisual modality subgroups exhibited low heterogeneity, whereas the visual modality subgroup showed high heterogeneity. No statistically significant differences were observed between the effect sizes of the three modalities. |
| Visual | 13 | P = 0.11 | -0.48 [ -1.19; 0.22] | 92% | |
| Auditory-Speech | 13 | | 0.30 [ 0.10; 0.50] | 16% | |
| Visual_Auditory-Speech | 8 | | 0.26 [0.03; 0.50] | 37% | |
| **Automation level** | | | | | The two level of automation did not appear to differentially affect the primary outcome, change in trust scores. Level 3 automation subgroup showed low heterogeneity among studies. The test for subgroup differences indicated that there is no statistically significant subgroup effect. |
| L3 | 12 | P = 0.30 | 0.17 [-0.06; 0.39] | 39% | |
| L5 | 21 | | -0.11 [-0.57; 0.36] | 87% | |
| **JBI quality** | | | | | Subgroup analysis of JBI quality did not appear to differentially affect the trust score. Low and moderate heterogeneity observed across studies with low and moderate JBI qualities, respectively. |
| Low | 2 | P = 0.54 | 0.28 [-0.06; 0.61] | 0% | |
| Moderate | 16 | | 0.10 [-0.22; 0.41] | 73% | |
| High | 16 | | -0.04 [-0.54; 0.46] | 88% | |
| **NDRT** | | | | | NDRT subgroup analysis result in a significant small positive effect of anthropomorphism on human trust in automated vehicles across studies that lacked NDRT. |
| No | 16 | P = 0.04 | 0.34 [0.17; 0.52] | 45% | |
| Yes | 9 | | 0.03 [-0.22; 0.27] | 17% | |
| **Role of participant** | | | | | When participants assumed the driver role in the studies, anthropomorphism had a significant small positive effect on human trust in automated vehicles, with low heterogeneity observed across studies. Although a moderate effect size was seen in the pedestrian group, this should be considered cautiously due to high heterogeneity and non-significant differences. |
| Driver | 22 | P = 0.15 | 0.26 [ 0.11; 0.41] | 37% | |
| Passenger | 3 | | 0.02 [-0.77; 0.81] | 76% | |
| Pedestrian | 9 | | -0.71 [-1.73; 0.30] | 94% | |
| **Driving simulation** | | | | | Anthropomorphism in driving simulator studies had a significant small positive effect on human trust in automated vehicles, while it did not appear to differentially affect on human trust in automated vehicles within real car, or virtual reality and video/picture simulation studies. The driving simulation subgroup analysis reduced heterogeneity across driving simulator and virtual reality studies. |
| Virtual Reality | 5 | P = 0.47 | -0.01 [-0.41; 0.39] | 58% | |
| Simulator | 17 | | 0.23 [0.04; 0.42] | 33% | |
| Real car | 4 | | -0.49 [-2.32; 1.33] | 90% | |
| Video/Picture | 8 | | -0.28 [-1.26; 0.70] | 94% | |

K: Number of studies; CI: Confidence interval.

Table 4. Meta-regression results for Age and Gender

| Covariate | Coefficient | Standard error | 95 % CI | *Z* value | *P* value |
|---|---|---|---|---|---|
| Average Age (K = 30) | 0.0309 | 0.0172 | [-0.0028; 0.0646] | 1.7990 | 0.0720 |
| Intercept | -0.9282 | 0.5372 | [-1.9812; 0.1247] | -1.7278 | 0.0840 |
| Female % (K = 31) | 0.0043 | 0.0117 | [-0.0185; 0.0272] | 0.3695 | 0.7118 |
| Intercept | -0.1894 | 0.5490 | [-1.2654; 0.8865] | -0.3451 | 0.7300 |

K: Number of studies; CI: Confidence interval.

### 2.5.Publication bias and sensitivity analysis

The funnel plots illustrated a roughly symmetrical distribution of the included studies, indicating no strong evidence of publication bias. These inspections were supported by the statistically non-significant findings of Egger's tests for small-study effects (Fig. 5; A ($p$ = 0.59), B ($p$ = 0.78), and C ($p$ = 0.10)).

The sensitivity analysis was conducted using a leave-one-out method for the meta-analysis comparing anthropomorphic and non-anthropomorphic interfaces. The finding revealed that removing the study by (Chung et al., 2024) (Study ID = 7) slightly increased the effect size (SMD = 0.17, 95% CI [0.02, 0.31], p = 0.02; $I^2$ = 60%), which considered as an influential factor (Fig. 6). However, the effect size remained trivial, suggesting that the difference in effect size was not considerable, and the overall result is robust. The sensitivity analysis for the meta-analysis of deep versus superficial anthropomorphism confirmed that no individual study contributed significantly to altering the significance of the result. This suggests that the overall result of this meta-analysis is robust. Finally, a sensitivity analysis was conducted to assess the robustness of the finding of the anthropomorphic interface versus baseline meta-analysis. The finding showed that the exclusion of Study ID = 9 (Du et al., 2019) increased the effect size to 0.23 (small effect, 95% CI: 0.08 to 0.38, $p$ = 0.002; $I^2$ = 7.1%, Fig. 8). Similarly, omitting Study ID = 58, (Wallbridge et al., 2024) resulted in an increased effect size of 0.20 (small effect, 95% CI [0.03, 0.37], $p$ = 0.023; $I^2$ = 33.1%, Fig. 8).

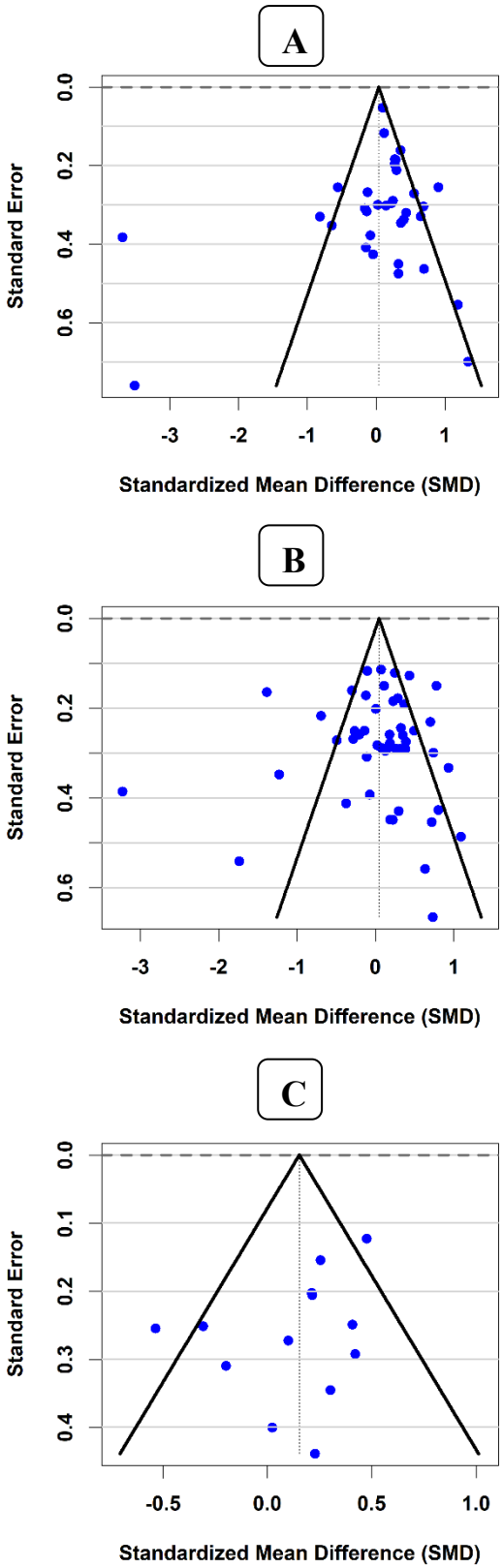



**Fig. 5.** Funnel plots of included studies for meta-analysis models (A) Anthropomorphic vs. non-anthropomorphic interfaces, (B) Deep vs. superficial anthropomorphism, and (C) Anthropomorphic vs. baseline. The funnel plots show the observed standardised mean differences (on the x-axis) against standard errors (on the y-axis).

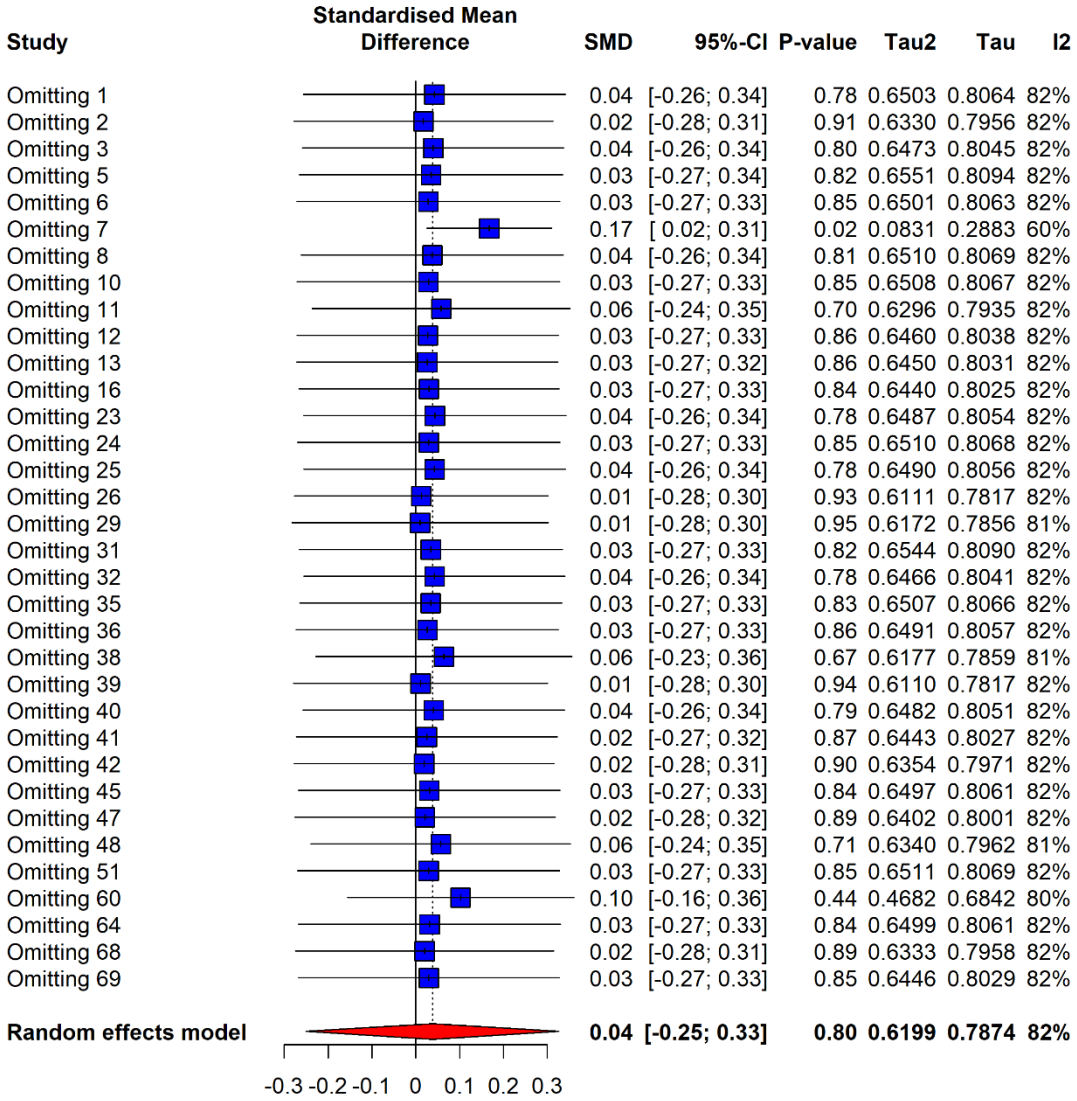



**Fig. 6.** Leave-one-out sensitivity analysis for the meta-analysis comparing anthropomorphic vs. non-anthropomorphic interfaces.

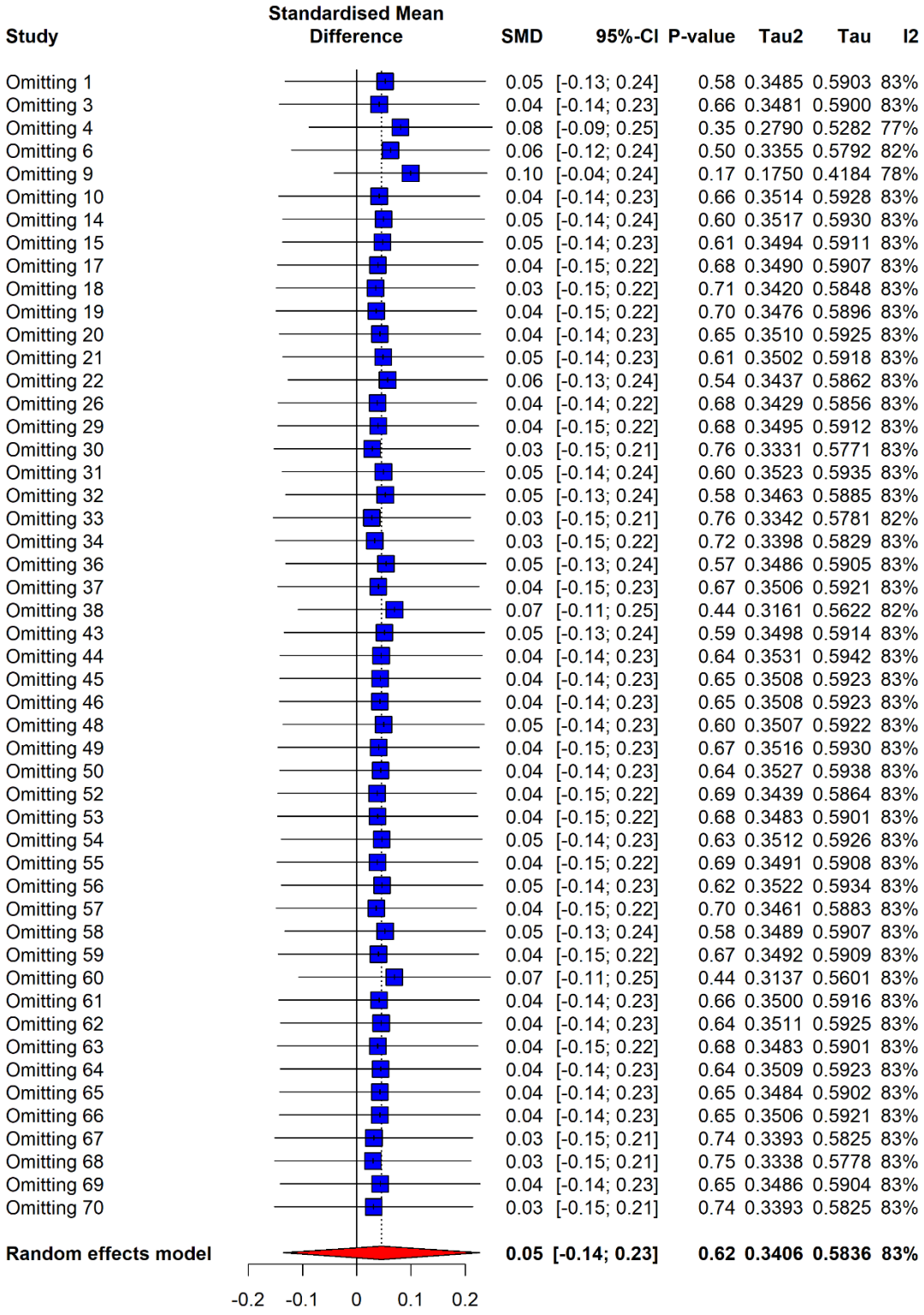



**Fig. 7.** Leave-one-out sensitivity analysis for the meta-analysis comparing deep vs. superficial anthropomorphism.

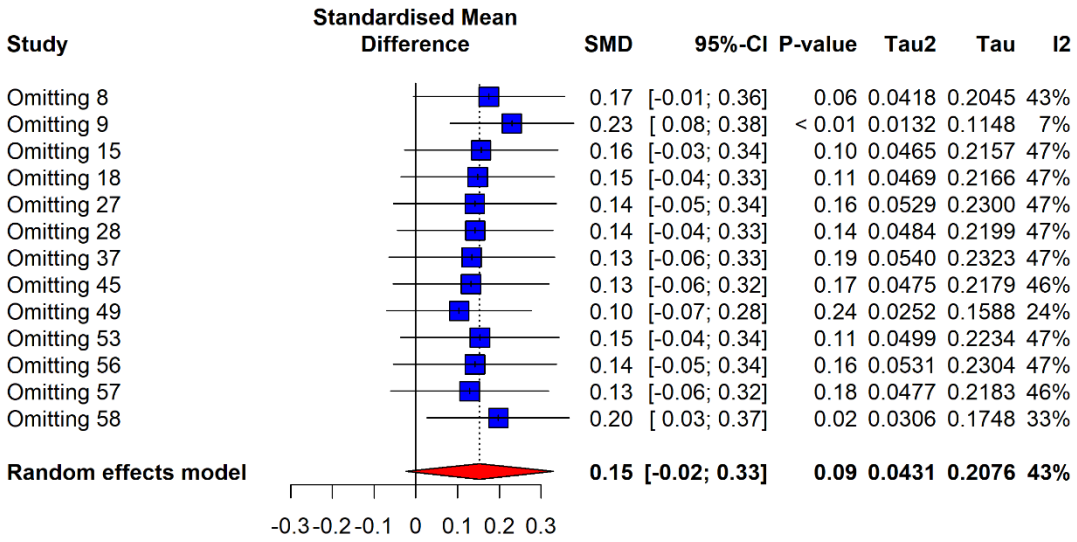



**Fig. 8.** Leave-one-out sensitivity analysis for the meta-analysis comparing anthropomorphic interface vs. baseline.

## 3. Discussion

This systematic review and meta-analysis investigated the impact of anthropomorphism interventions on trust in AVs. Authors answered the question of whether attributing human-like characteristics to AVs (i.e., anthropomorphizing them), genuinely enhances user trust.

**The main finding of the meta-analysis indicated that anthropomorphic interfaces did not produce a statistically significant improvement in trust toward AVs when compared to either non-anthropomorphic interfaces or the absence of any interface.**

The included studies reported both positive and negative effects of anthropomorphism on trust. Therefore, the pooled outcome of this meta-analysis is not unexpected. However, the finding of this meta-analysis does not align with those of similar meta-analyses conducted outside the domain of AVs such as in the field of HRI. A meta-analysis by (Roesler et al., 2021) reported a positive effect of anthropomorphism on trust to robots with a medium effect size (d = 0.726). Also, a meta-analysis by Blut et al. (2021) investigated the relationship between robot anthropomorphism and trust, based on 24 effect sizes from a cumulative sample of 2,365 participants. The results revealed a small but significant positive correlation (r = 0.19, 95% CI [0.00, 0.38]) across studies involving physical robots, chatbots, and other automated systems in service contexts (Blut et al., 2021). Other meta-analysis has also examined outcomes beyond trust, such as user experience with anthropomorphic interfaces. In this case, as well, the overall

estimates have indicated positive effects of anthropomorphism. A meta-analysis by Yee et al. (2021) examined 25 studies comparing interfaces featuring visually embodied, human-like agents with those without such agents. The results indicated a small but statistically significant effect size ($r = 0.13$, $p < .001$) in favor of interfaces with human-like representations (Yee et al., 2007).

For discussing the misalignment between paper's results and other studies outside the AV domain, authors point to inherent differences between human-AV interaction and HRI, the difference between results is reasonable. Inherent differences include the typically open setting of HRI versus the enclosed environment of human–vehicle interaction, as well as the motion and inherent risks of vehicles compared to the stationary nature of most robot interactions.

However, the reasons for ineffectiveness of anthropomorphic interfaces in enhancing trust in AVs can be explored through the following considerations (*theoretical considerations + technical view to trust + Confounding variables such as initial trust do+ Interaction time + Perceived anthropomorphism + some other influential parameters*).

Despite of previous theories, there are also several social and psychological theories have explored the potential negative consequences of anthropomorphizing technology such as the Uncanny Valley Theory (Burleigh et al., 2013; Ho & MacDorman, 2010), Expectancy Violation Theory (Bevan et al., 2014), and Media Equation Theory (Złotowski et al., 2018). These frameworks emphasize that when an anthropomorphized object fails to replicate human physical, behavioral, or social cues adequately, it is likely to fall short of user expectations and result in diminished trust. Our results confirmed the ineffectiveness of anthropomorphic design in enhancing users' trust in AVs. One possible justification is that if a vehicle, despite being designed with human-like features, fails to meet the expectations, anthropomorphism proves ineffective.

From the technical point of view, this issue can also be interpreted. Trust is a multidimensional construct, and the quality and accuracy of the vehicle's technical performance play a crucial role and perhaps it is more important than the content of the interfaces (anthropomorphic or normal or even without any interfaces). A non-accurate AV, whether equipped with anthropomorphic interfaces or not, generally diminishes trust in the vehicle. In such cases, the effect of anthropomorphic interfaces is masked and does not function as intended in their design.

Moreover, other influential parameters such as initial trust, which were not addressed in this particular study, have been shown in previous research to significantly affect dynamic trust (Niu et al., 2018; Yuan & Yu, 2024). This means that individuals with initially low trust will maintain low dynamic trust (Manchon et al., 2022) regardless of whether the autonomous vehicle is anthropomorphized or not, and this may influence the relationship between anthropomorphism and trust. So, controlling such confounding parameters could produce different results which was impossible in the current study due to lack of reports by the included studies.

Generally, it is widely believed that trust in a vehicle much like the process by which trust develops between humans, is a medium- to long-term process (Glikson & Woolley, 2020; Stilgoe, 2023; Walker et al., 2023). In contrast, most of the studies included in this meta-analysis involved only brief exposure to the scenario, typically lasting a few minutes and at most up to some limited hours. Therefore, short-term interactions may not necessarily lead to substantial changes in overall trust in the AV.

Additionally, there are instances in which, despite researchers' efforts to design anthropomorphic features into the experimental setup, participants' perceptions of anthropomorphism (i.e., perceived anthropomorphism (Li et al., 2023; Tsfasman et al., 2021)) do not align with the intended design, ultimately resulting in a failure to achieve the intended manipulation. In 2023, to examine the distinction between designer-intended and user-perceived anthropomorphism in human–AV interaction, it was found that, contrary to the authors' hypothesis, adding visual features did not result in higher user-perceived anthropomorphism (Lawson-Guidigbe, Amokrane-Ferka, et al., 2023b).

Taken together, the above-mentioned 6 aspects may help explain the limited effectiveness of anthropomorphic interventions in enhancing trust to AVs.

**Pooled data analysis revealed that variations in the depth of anthropomorphism (superficial vs. deep) results in a trivial effect in users' trust in AV**

The concept of superficial versus deep anthropomorphism has been applied in various domains, including HRI, and its influence on trust in both vehicles and other technologies has been documented (Carter et al., 2024; Cheng et al., 2022; Ratajczyk, 2022).

Based on Uncanny Valley theory, many studies emphasize that lower levels of anthropomorphism tend to have a more positive effect on trust than higher levels. The results

of a study showed that human-like social interactions in AVs can enhance trust and acceptance, but overuse of anthropomorphic features (e.g., deep anthropomorphism) such as addressing the driver by name and expressing emotions may have counterproductive effects. These findings highlight the potential trade-offs in designing social AV interactions, reflecting the "uncanny valley" effect (Kim et al., 2025). Contradictory findings have also been reported. The argument is that deep anthropomorphism can foster stronger emotional, cognitive, and social connections between humans and technology, making it more effective than superficial anthropomorphism (Carter et al., 2024). In the current study our analysis went beyond the previous dual and our analysis showed that there are no differences between superficial and deep anthropomorphism, at least regarding trust to AVs.

Therefore, the use of anthropomorphic features must be approached with caution, and some studies have even emphasized their context-specific nature (Roesler et al., 2022). In the AV domain (Park et al., 2024a) this implies that when long-term interaction is expected such as in the case of a privately owned AV over an extended period, deep anthropomorphism may be more appropriate (Guo et al., 2025; Wu et al., 2023b). In contrast, in short-term interactions, such as those involving public autonomous shuttles or shared rides where the risk of over-trust can be harmful, the use of superficial anthropomorphism is recommended (Wu, 2024).

**In the included studies, anthropomorphism interventions were implemented in three different forms: auditory, visual, and combined auditory-visual. Subgroup analysis based on the modality of anthropomorphism revealed that auditory anthropomorphic interfaces (characterized by human voice features) produced a small positive in trust toward AVs, whereas visual anthropomorphic interfaces did not.**

Anthropomorphism has been implemented through various modalities (Bai et al., 2024; Burgoon et al., 2000). However, it is essential to consider the context of interaction, that is, the setting in which human-technology interaction takes place when discussing the modality through which anthropomorphism is conveyed. For example, given that driving (both manual and almost all automated levels except L4 and L5) is inherently a cognitively demanding process centered on the visual modality, and that users generally perceive driving as a multimodal task with a strong emphasis on vision (Mosaferchi et al., 2025; Pauzié, 2008; Zakerian et al., 2018), it follows that anthropomorphic voice (engaging the auditory modality) and anthropomorphic visuals (engaging the visual processing in the brain) may have different effects.

At the very least, this distinction has been debated and examined for more than half a century in various studies (Wickens, 2008), with a general consensus suggesting that when one modality is heavily engaged, it is preferable for supplementary information to be delivered through another modality (S. Kim et al., 2024; Nason & Wilbiks, 2025). This approach helps avoid excessive workload, reduced situational awareness, or even safety issues.

**Studies conducted in simulator environments demonstrated that anthropomorphic interfaces significantly increased trust in AVs, whereas this effect was not observed in field studies, including test-track driving tests or wizard-of-OZ.**

Researchers have recognized for several decades that certain parameters cannot be tested in real-world environments due to safety and or ethical risks and or time/budget constraints (Silva et al., 2024). To address this, simulators were introduced and have since been widely used (in case of driving simulator, the first study was in 1934 (Caird & Horrey, 2011) and for flight training was 1910 (Bruck et al., 2021)). While offering certain advantages, the limitations inherent in simulator studies cannot be ignored. These include issues such as overtraining (Weitz & Adler, 1973), motion sickness (Igoshina et al., 2022), and even the potential for exaggerations of results.

After reviewing the benefits and positive aspects of driving simulators, Xuemin Hu et al. discussed the "*reality gap*" challenge that lies in effectively transferring driving knowledge from the virtual simulation world to the reality world. They believed that this gap arises due to differences in lighting, textures, vehicle dynamics, and agents' behaviors between the two environments (X. Hu et al., 2024).

Most studies in this area were conducted using driving simulators. However, several limitations should be noted. First, it is uncertain whether driver behavior in a simulator truly reflects behavior in real-world driving situations, which raises concerns about the generalizability of findings (Caird & Horrey, 2011; Hock et al., 2018). Real driving scenarios or actual road conditions may evoke different perceptions and reactions compared to simulated environments. Second, a systematic review has shown that, despite the safety of simulator studies, no firm conclusions can be made about the efficacy of training programs or interventions when tested only in simulated settings (Martín-delosReyes et al., 2019). Finally, trust in automated systems may develop more quickly in virtual reality or simulator contexts due to the absence of real-world risks, which could mask how such trust would evolve in genuine driving situations.

Taken together, these issues suggest that conclusions drawn from simulator-based studies should be interpreted with caution.

**Performing or not performing NDRTs had a significant impact on the relationship between anthropomorphic interfaces and trust in AVs; specifically, this relationship was non-significant when drivers were engaged in NDRTs, whereas it was significant when drivers were not performing these secondary tasks.**

With the introduction of high-level AVs, such as L3 and above, the interaction pattern between driver and vehicle fundamentally changes. Users are now able to engage freely in secondary activities unrelated to driving, including holding work meetings, eating, sleeping, using mobile devices, and more (Cai et al., 2024; Weigl et al., 2023). Some studies assess trust in AVs by observing participants' voluntary engagement in NDRTs during automated driving. When individuals are given the choice to perform or ignore an NDRT, the duration and ease with which they shift their attention away from the driving environment is interpreted as an indirect indicator of trust (Körber, Baseler, et al., 2018; Pan et al., 2023).

Although Zhang et al. studied anxiety reduction and risk perception instead of trust, their findings showed that auditory anthropomorphic explanations were more effective than visual but less anthropomorphic explanations, especially for high-suspicion individuals without a secondary task. This result supports the broader link between voice-based anthropomorphism and user trust in AV (Q. Zhang et al., 2023). This result is in-line with the current study as both of them implied that, doing NDRTs can decrease the effectiveness of anthropomorphic explanation. Similarly, Hong et al. also implied that speech explanations increased trust when passengers were not engaged in NDRTs (Hong et al., 2025).

Overall, these findings are also theoretically supported by common cognitive theories, such as multiple resource theory and or information process theory (Wickens, 2008; Wickens & Carswell, 2021). These theories suggest that when people are engaged in another cognitive task, they are less likely to process additional messages, whether anthropomorphic or not. As a result, their trust in AV may be influenced by doing NDRTs. So, it can be concluded that in the case of stressful/high-load situation, even some positive effect of anthropomorphic interfaces could be diminished.

**The effect of anthropomorphic interfaces on trust in AVs differed between drivers and pedestrians (in-vehicle interfaces vs. eHMIs). The effect was significant among drivers but not significant among pedestrians.**

Generally, eHMIs are seen as an effective way to communicate when an AV has no driver or the driver is busy with non-driving tasks and cannot support pedestrians in making the best decision in interacting with the AVs (Brill et al., 2023).

When comparing drivers and pedestrians in their responses to anthropomorphic interfaces, a key difference is the duration of exposure. Drivers may interact with in-vehicle interfaces for extended periods, while pedestrians typically see an eHMI for only a few seconds. Undoubtedly, when the exposure time to a message is short, the message must be conveyed quickly and accurately (Eisma et al., 2023). In such situations, anthropomorphizing the message may serve no real benefit other than adding cognitive load for the pedestrian, who may not have sufficient time to properly interpret it (Gong). In describing the differences between anthropomorphic and non-anthropomorphic eHMIs in terms of message-processing speed, some studies have distinguished between egocentric and allocentric messages (Eisma et al., 2021). Allocentric messages convey the AV's intentions or upcoming actions, requiring the road user to interpret the message from the vehicle's perspective, which takes more time to act upon (*e.g., "Car will not stop, you cannot pass"*). In contrast, egocentric messages directly instruct the road user on what action to take, offering a clear and immediate call to action (*e.g., "I am approaching, you cannot pass"*) (Bazilinskyy et al., 2019; Brill et al., 2023).

Mentioned by Man et al. in their bibliometric analysis (Man et al., 2025), many current eHMI studies rely mostly on virtual reality and video-based simulations, which do not fully capture the complexity of real-world environments. Similar to the concerns previously raised about driving simulators, these experimental setups may limit the ecological validity of findings. Therefore, caution is needed when interpreting the results.

**Overall, older users exhibited a greater benefit of anthropomorphic interfaces to enhance trust in AVs.**

The meta-regression analysis indicated a marginally significant effect of participants' average age on human trust in AVs. Specifically, for every 10-year increase in age, the SMD of trust increased by approximately 0.3 units. This trend is consistent with the findings of the sensitivity analysis (Fig. 6), which showed that excluding a study focused specifically on children resulted in a higher overall effect size of anthropomorphism on trust in AVs.

It is important to note that older individuals are generally less likely to be interested in adopting AVs and tend to have lower initial trust compared to younger users (Lajunen & Sullman, 2021; Robertson et al., 2019). Therefore, this reflects a multi-layered relationship between user age,

initial trust, dynamic trust, and the perceived need for anthropomorphic information to enhance trust (at least proven in other technologies and also a possibility in AVs) (Pak et al., 2014).

Due to natural eyesight and hearing declines associated with aging (Mosaferchi & Naddeo, 2025), older adults have a greater need for driver-assistance in-vehicle technologies (Eby et al., 2018). Regarding anthropomorphic driving assistance system, a recent study by Li et al. (2025), proved that, systems with higher anthropomorphic features significantly improved user trust in elderly AI navigation users (Li et al., 2025).

Outside the AV domain, older people have higher satisfaction with companion robots with higher levels of anthropomorphism, and anthropomorphism has a significant positive effect on their satisfaction (Liang et al., 2025). Sinnema et al. (2019) evaluated short-term interactions with a humanoid social robot among two age groups - elderly participants and students. While both groups reported a generally positive attitude toward the robot, elderly participants rated the robot as significantly more useful than the younger group. This difference was attributed to varying needs and expectations, suggesting that older adults may derive greater value from anthropomorphic systems (Sinnema & Alimardani, 2019). These findings align with our meta-analysis, which also indicated stronger positive responses to anthropomorphic interfaces among older users. However, it is important to note that Sinnema et al.'s study did not assess trust, and the context involved social interaction rather than AV. Thus, while supportive, these results reflect a different dependent variable and application domain. In the study by Tewari and Lindgren, the concept of anthropomorphism is not the central focus but is subtly embedded throughout the discussion of socially intelligent robots (Tewari & Lindgren, 2022). While the term "anthropomorphism" is mentioned briefly in reference to robots that evoke social responses, most of the participants' feedback reflects implicit anthropomorphic interpretations such as referring to the robot as "he", comparing it to a friend or family member, and expecting it to recognize and respond to human emotions. These expectations point to an underlying attribution of human-like traits, particularly among older adults, who valued empathy, politeness, and social appropriateness more than younger users. These findings align with our meta-analysis, which showed that older individuals tend to respond more positively to anthropomorphic systems.

Conversely, Letheren (2016) found that younger, single, more creative, and conscientious individuals are more receptive to anthropomorphic messages, suggesting that anthropomorphic tendency is shaped by personal traits such as age, personality, and experiential thinking

(Letheren et al., 2016). This is in contrast with our meta-analysis that showed older adults responding more positively to anthropomorphic interfaces in AV contexts. A likely reason for this difference is the variation in context and dependent variables: Letheren's study focused on general anthropomorphic tendency, while our research examined trust within a safety-critical domain. Also, in contrast to our meta-analysis, which suggested that older adults may benefit more from anthropomorphic interfaces in AVs, Pak et al. found that younger adults showed significantly higher trust in anthropomorphic aids than in non-anthropomorphic ones, whereas older adults' trust remained unaffected by anthropomorphic design (Pak et al., 2012).

**No significant gender differences were observed in the relationship between anthropomorphism and trust in AVs.**

When considering the role of gender in the use of anthropomorphic interfaces within AVs, it is important to recognize that - similar to age- gender independently influences both trust in the vehicle (Modliński et al., 2022) and perceived anthropomorphism (Crowell et al., 2019).

Although it is commonly believed that anthropomorphizing technology is generally more enjoyable or appealing to women, some studies have shown that men may actually process anthropomorphic information with greater sensitivity than women (Pelau et al., 2022). Some studies, after controlling for participants' age, have reported that the tendency to anthropomorphism is independent of gender, suggesting that there is no significant difference between men and women in their perception of anthropomorphism (Letheren et al., 2016).

However, it is important to note that the context of interaction with an anthropomorphic object or interface plays a crucial role in shaping user perceptions. To more accurately interpret and generalize the findings of this study, attention should be directed toward contexts that closely resemble driving scenarios such as interaction during typical-speed driving, within the confined space of a vehicle cabin, or in a driving simulator replicating in-vehicle conditions. A recent study (Lee & Cho, 2024) examining the interplay between gender and anthropomorphic tendency in the context of AVs revealed that user perceptions vary significantly based on this interaction. Specifically, men with a high propensity to anthropomorphism reported more positive attitudes and higher competence evaluations of AVs, whereas women with similar tendencies exhibited more negative perceptions and rated the vehicles as less competent. These findings underscore the importance of considering the combined effect of gender and anthropomorphic tendency in the design and evaluation of in-vehicle interfaces.

## 4. Limitations, practical implications and future directions

The findings of this meta-analysis offer several actionable insights for stakeholders in AVs development. For researchers, the results highlight the need to move beyond the assumption that anthropomorphic design inherently increases trust; future studies should examine the conditions under which such designs are effective, with particular attention to modality, user demographics, and driving context. For interface designers, investing in auditory anthropomorphic cues may provide modest trust benefits, especially for older users, whereas relying solely on visual cues may not yield the same effect. For manufacturers, these insights suggest that incorporating anthropomorphic elements should be targeted rather than universal tailored to user groups, operational environments, and usage scenarios.

Although the present meta-analysis aimed to control for methodological limitations as much as possible, certain constraints remained. For instance, due to the limited assessment of trust in one of the included studies, focusing solely on behavioral trust without reporting subjective trust this study was excluded despite initial eligibility (Hester et al., 2017). It is therefore recommended that future meta-analyses incorporate both behavioral and subjective measures of trust to provide a more comprehensive evaluation.

## 5. Conclusion

By aggregating findings from 66 studies, this meta-analysis provides a nuanced picture of how anthropomorphic interfaces can shape trust in AVs. The overall pattern suggests that adding human-like elements to interfaces does not automatically translate into greater trust, and differences in the depth of anthropomorphism was not significant regarding trust. The form in which anthropomorphic cues are delivered matters: Voice-based designs demonstrated modest potential, whereas purely visual cues did not yield significant effects. Individual and situational factors further shape these outcomes. Older participants tend to benefit more from anthropomorphic features, yet trust levels do not differ notably between men and women in related to their response to anthropomorphism interventions. In response to anthropomorphic interfaces, trust in simulated AVs in simulators were stronger than real-world trials, and the driver's engagement in secondary activities can diminish the effect of anthropomorphic interventions on trust. The intended user group also matters and drivers respond more positively to in-vehicle anthropomorphic cues than pedestrians to external HMIs.

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**Declaration of generative AI and AI-assisted technologies**: In the final preparation of this manuscript, Chat GPT-5 is used to check the grammar and increase the readability of the content. After that, all the content is re-checked by the corresponding author to ensure about its intended meaning.

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